

Coffee Shop Sales Data Analysis: Insights and Dashboards



An Excel Portfolio Project
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Project Overview

- Objective: Analyze Coffee Shop sales data to uncover trends and provide actionable insights.
- Key Analysis Areas:
 - * Sales by weekdays.
 - * Sales by product category.
 - * Sales by Individual product.
 - * Category Distribution by Sales.
 - * Quantity Ordered Based on hours.
 - * % size Distribution based on hours.
 - * Top 5 Products based on sales.





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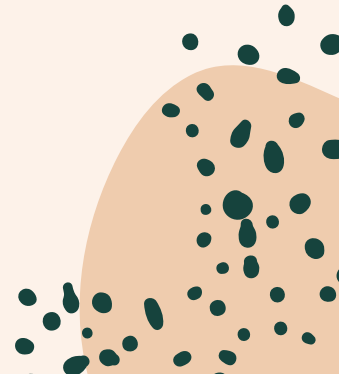
Data Source: Coffee Shop Data Sales dataset from Maven Analytics.

Data Size: 149117 transactions over six months.

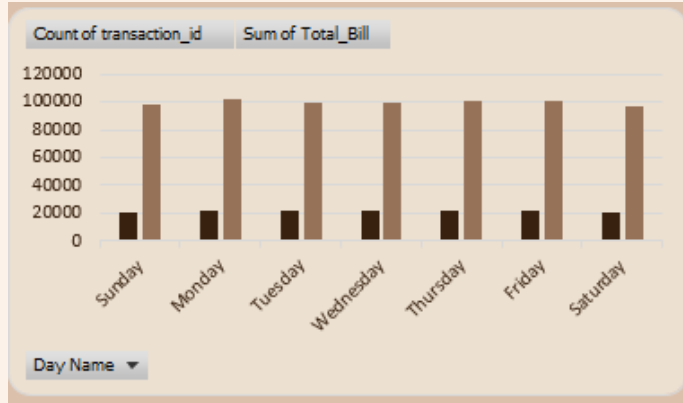
Preparation Steps:

- * Cleaned missing and inconsistent entries.
- * Formatted Data and time for Analysis.
- * Added a new columns: hour, month name, total_bill, Weekday.
- * Categorized Products into groups.

Tools Used: Microsoft Excel(pivot tables, pivot charts, formulas like VLOOKUP, power pivot analyze, slicers and dashboards.



Analysis 1: Sales by Weekdays

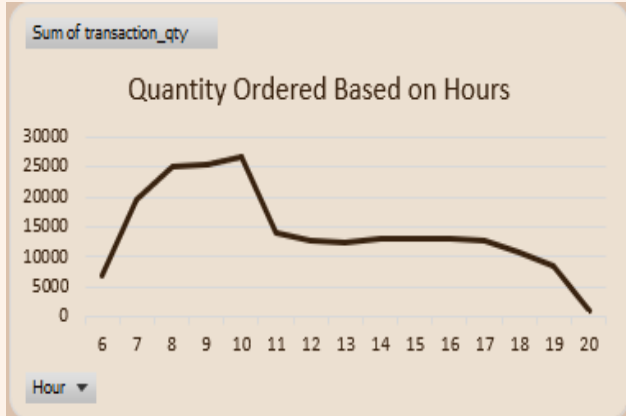


Objective: Identify which days of the week generate the highest sales.

Method: Used pivot tables to aggregate sales by day of week.

Key Findings: Monday and Friday are the busiest days contributing most to the sales.

Analysis 2: Quantity Ordered Based on Hours

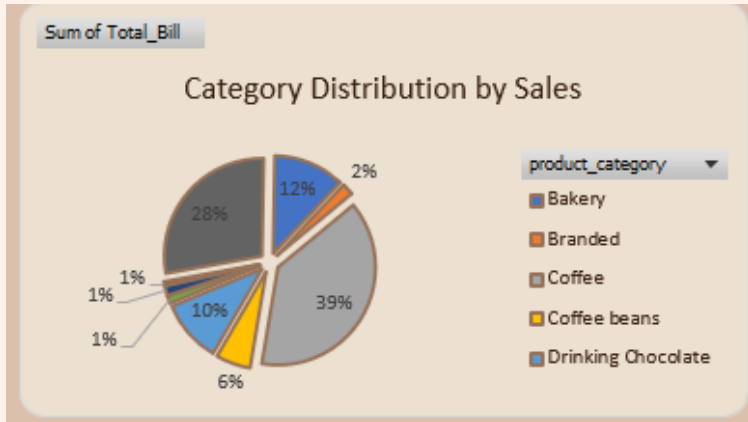


Objective: Determine How quantity ordered Varies by hour.

Method: Added a new Column Hour in power query based on time and analyzed using Pivot table.

Key Findings: Quantities of coffee ordered were peaked during 8am to 10am.

Analysis 3: Determine How Sales vary by Category Basis



Objective: Identify which Product Category Contributed most to the sales.

Method: Used Pivot table to analyze data and ranked them highest to lowest.

Key Findings: Coffee category Product Contributed most to the sales.

Analysis 4: Footfall and Sales over Various Sales Location

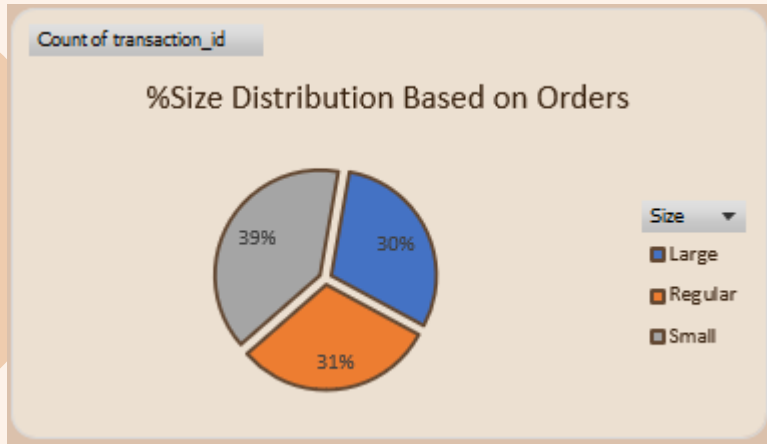


Objective: Identify How footfall and Sales vary by Store Location.

Method: Ranked Product by Sales revenue and Count of transaction_id using Excel sorting and filtering Features and charts for visualization.

Key Findings: Hell's Kitchen Contributed most to the footfall and sales revenue.

Analysis 5: Percentage Size distribution Based on orders

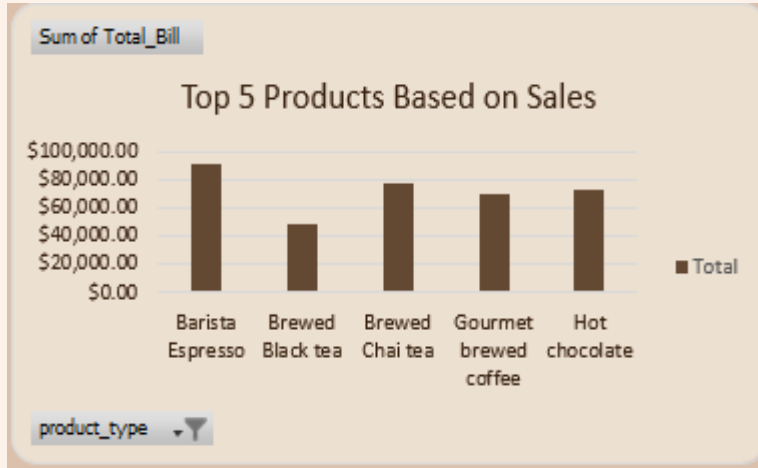


Objective: Determine the Percentage Size distribution Based on orders.

Method: Grouped Sales Based on size and used Pivot table to analyzed Data.

Key Findings: Large size Cup was the most ordered in coffee sales.

Analysis 6: Top 5 Products Based on Sales



Objective: Identify top 5 Products based on sales.

Method: Ranked Product by Sales Volume and revenue and used pivot table for analyzing and creating Dashboards.

Key Findings: Barista Espresso was the most ordered coffee followed by Brewed Black tea, Brewed Chai tea, Gourmet Brewed Coffee, Hot Chocolate.

Final Dashboard

Objective: Summarize all insights in a single, interactive view for decision-making.

Components:

- Top 5 Products Based on Sales revenue. (Bar Chart)
- Quantity Ordered Based on hours. (Line Chart)
- Sales and order distribution by weekdays. (Bar Chart)
- % size distribution based on orders. (Pie Chart)
- Footfall and Sales over Store Location. (Bar Chart)
- Category Distribution by Sales. (Pie Chart)
- Slicers by Month Name and Day Name.

Final Dashboard

Coffee Shop Sales



\$698,812.33
Total Sales

149116
Total FootFall

\$4.69
Avg Bill/Person

1.44
Avg Order/Person

Month Name

January

February

March

April

May

June

Sum of transaction_qty

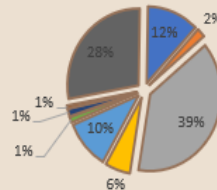
Quantity Ordered Based on Hours



Hour

Sum of Total_Bill

Category Distribution by Sales



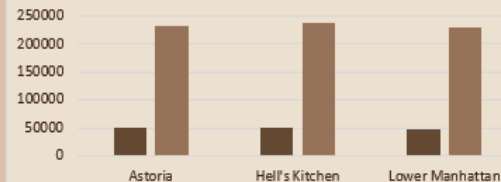
product_category

- Bakery
- Branded
- Coffee
- Coffee beans
- Drinking Chocolate

Count of transaction_id

Sum of Total_Bill

Footfall and Sales over Store Location



store_location

Day Name

Sunday

Monday

Tuesday

Wednesday

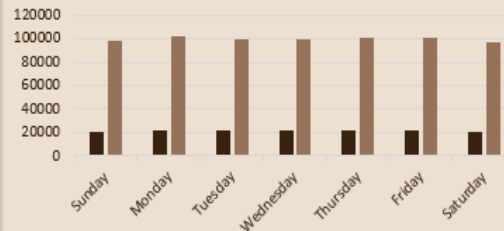
Thursday

Friday

Saturday

Count of transaction_id

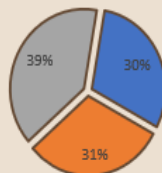
Sum of Total_Bill



Day Name

Count of transaction_id

%Size Distribution Based on Orders

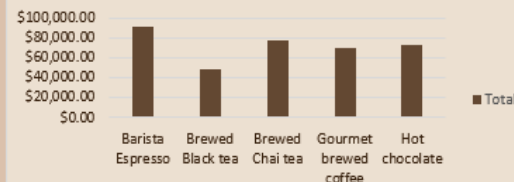


Size

- Large
- Regular
- Small

Sum of Total_Bill

Top 5 Products Based on Sales



product_type

Conclusions and Next Steps



Conclusions: This Project demonstrates the power of Excel in analyzing Sales Data to Drive Business Decisions.

Skills Showcased: Data Cleaning, Pivot tables, Power Query, Power pivot, Charting, Dashboard Creation.

Next Steps:

- Expand Analysis to include customer demographics.
- Integrate Predictive Analytics to forecast future sales.